Sri Lanka Institute of Information Technology

Game Development - SE4031

Assignment 1

Registration Number	Student Name
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Frostfire

Game Title

Frostfire

Summary of the Game Concept

Frostfire is a 2D side-scrolling interactive game developed using Unity. In this game, the player controls a dragon that continuously flies through a dangerous environment. The objective is to survive as long as possible by avoiding obstacles, defeating enemies, and collecting health packs. The dragon can shoot fire projectiles to destroy enemies while avoiding incoming ice spear attacks from Night King enemies. The difficulty increases over time as the speed of the game gradually increases.

Core Features Implementation

1. Health System

- The player's health is displayed as a UI health bar.
- Health decreases when the dragon collides with:
    - Obstacles
    - Enemies (Night King)
    - Ice spear projectiles
- The dragon continues moving even after collisions and only stops when health reaches zero.
- The health value is limited by a maximum value.
- Health resets to maximum when the game is restarted.

2. Score System

- The score is displayed on the UI.
- The score increases when the player successfully passes through the gap between obstacles.
- Additional score can be gained by destroying enemies using fire projectiles.
- The score resets to zero when the game restarts.

3. Health Packs

- Health packs appear randomly in the game.
- They are collidable objects.
- When collected:
    - The player's health increases (up to maximum limit).
    - A sound effect is played.
    - The health pack is destroyed.
- Health packs are also destroyed automatically when they move off-screen.

4. Player Projectile Attack

- The player can shoot fire using the left mouse button.
- Fire projectiles:
    - Spawn from the dragon's fire point.
    - Travel toward the mouse direction.
    - Destroy enemies upon collision.
    - Are destroyed after collision or after a short time.

5. Enemy Mechanics

- Enemies (Night King) spawn from the right side.
- They move toward the player.
- Enemies:
    - Throw ice spear projectiles toward the dragon.
    - Damage the player on collision.
    - Are destroyed when hit by fire projectiles.
    - Increase the player's score when destroyed.
    - Are automatically removed when they go off-screen.
- Enemies are visually different from health packs.

6. Speed Increase Feature

- The game speed gradually increases over time.
- This makes obstacles, enemies, and objects move faster.
- The increase is smooth and balanced to maintain gameplay difficulty.

Screenshots of gameplay

Score: 33

Score: 68

Creative Feature

The game includes a dynamic visual effect where the health bar color changes based on the player's health:

- High health → Warm/fire color (orange/red)
- Low health → Cold/ice color (blue)

Additionally, the dragon includes a wing animation to simulate flying, and sound effects are added for shooting fire and collecting health packs, improving the overall game experience.

Control Guide

Action	Control
Move Up	W / Up Arrow
Move Down	S / Down Arrow
Move Left	A / Left Arrow
Move Right	D / Right Arrow
Shoot Fire	Left Mouse Click

Credits

- Game developed using **Unity Engine**
- Audio and assets used are either self-created or sourced from free resources
- Tools used:
    - Unity Editor
    - Visual Studio
    - Adobe Photoshop
    - Envato

Video Link

Google drive link